

APO-FLUVOXAMINE

Fluvoxamine Maleate Tablets 50 mg

Antidepressant, Antiobsessional Agent

Pharmacology

The antidepressant and antiobsessional actions of fluvoxamine are believed to be related to its selective inhibition of presynaptic serotonin reuptake in brain neurons.

There is minimum interference with noradrenergic processes, and in common with several other specific inhibitors of serotonin uptake, fluvoxamine has very little in vitro affinity for α_1 , α_2 , β_1 , dopamine₂, histamine₁, serotonin₁, serotonin₂ or muscarinic receptors.

Pharmacokinetics: APO-FLUVOXAMINE is entirely absorbed following oral administration. Maximum plasma concentrations occur within 3-8 hours of dosing. After a single dose and slightly longer (17-22 hours) during repeated dosing, the mean plasma half-life is approximately 13-15 hours. Within 10-14 days, steady-state plasma levels are usually achieved.

APO-FLUVOXAMINE goes through an extensive hepatic conversion, mostly via oxidative demethylation, into at least 9 metabolites which are excreted by the kidneys. The two key metabolites showed insignificant pharmacological activity. In vitro binding of APO-FLUVOXAMINE to human plasma proteins is about 80%.

Indication

Major Depressive Episode. Obsessive Compulsive Disorder.

Recommended Dose

Depression

Adults: The recommended dose is 100 mg daily. Patients should start on 50 or 100 mg, given as a single dose in the evening. Dosage should be reviewed and adjusted if necessary within 3 to 4 weeks of initiation of therapy and thereafter as judged clinically appropriate. Although there may be an increased potential for undesirable effects at higher doses, if after some weeks on the recommended dose insufficient response is seen some patients may benefit from having their dose increased gradually up to a maximum of 300 mg a day. Doses up to 150 mg can be given as a single dose, preferably in the evening. It is advisable that a total daily dose of more than 150 mg is given in 2 or 3 divided doses. Dosage adjustments should be

made carefully on an individual patient basis, to maintain the patients at the lowest effective dose.

Patients with depression should be treated for a sufficient period of at least 6 months to ensure that they are free from symptoms.

Children/adolescents: APO-FLUVOXAMINE should not be used in children and adolescents under the age of 18 years for the treatment of major depressive episode. The efficacy and safety of APO-FLUVOXAMINE have not been established in the treatment of paediatric major depressive episode.

Obsessive Compulsive Disorder

Adults: The recommended dose is between 100-300 mg daily. Patients should start at 50 mg per day. Although there may be an increased potential for undesirable effects at higher doses, if after some weeks on the recommended dose insufficient response is seen some patients may benefit from having their dose increased gradually up to maximum of 300 mg a day. Doses up to 150 mg can be given as a single dose, preferably in the evening. It is advisable that a total daily dose of more than 150 mg is given in 2 or 3 divided doses. If a good therapeutic response has been obtained, treatment can be continued at a dosage adjusted on an individual basis. While there are no systematic studies to answer the question of how long to continue fluvoxamine treatment, OCD is a chronic condition and it is reasonable to consider continuation beyond 10 weeks in responding patients. Dosage adjustments should be made carefully on an individual patient basis, to maintain the patients at the lowest effective dose. The need for treatment should be reassessed periodically. Some clinicians advocate concomitant behavioral psychotherapy for patients who have done well on pharmacotherapy. Long-term efficacy (more than 24 weeks) has not been demonstrated in OCD.

Children/adolescents: In children over 8 years and adolescents there is limited data on a dose of up to 100 mg bid for 10 weeks. The starting dose is 25 mg per day. Increase every 4-7 days in 25 mg increments as tolerated until an effective dose is achieved. The maximum dose in children should not exceed 200 mg/day. It is advisable that a total daily dose of more than 50 mg should be given in two divided doses. If the two divided doses are not equal, the larger dose should be given at bedtime.

Mode of Administration: Oral route.

Contraindication

APO-FLUVOXAMINE is contraindicated in combination with monoamine oxidase inhibitors (MAOIs).

Treatment with fluvoxamine can be initiated:

- two weeks after discontinuation of an irreversible MAOI, or
- the following day after discontinuation of a reversible MAOI (e.g. moclobemide).

At least one week should elapse between discontinuation of fluvoxamine and initiation of therapy with any MAOI. Hypersensitivity to the active substance or to any of the excipients.

Precautions/Warnings

Suicide/suicidal thoughts: Depression is associated with an increased risk of suicidal thoughts, self-harm and suicide. This risk persists until significant remission occurs. As improvement may not occur during the first few weeks or more of treatment, patients should be closely monitored until such improvement occurs. It is general clinical experience that the risk of self-harm is highest shortly after presentation and the risk of suicide may increase again in the early stages of recovery. Furthermore, antidepressants may rarely increase the risk of suicidal thoughts and self-harm.

Other psychiatric conditions for which fluvoxamine is prescribed can also be associated with an increased risk of suicide-related events. These conditions may be co-morbid with major depressive disorder. The same precautions observed when treating patients with major depressive disorder should therefore be observed when treating patients with other psychiatric disorders. Patients with a history of suicide-related events, those exhibiting a significant degree of suicidal ideation prior to commencement of treatment are at a greater risk of suicidal thoughts or suicide attempts, and should receive careful monitoring during treatment. In addition, there is a possibility of an increased risk of suicidal behaviour in young adults.

Patients and caregivers of patients should be alerted about the need to monitor for the emergence of such events and to seek medical advice immediately if these symptoms present.

Akathisia/ psychomotor restlessness: The use of fluvoxamine has been associated with the development of akathisia, characterised by a subjectively unpleasant or distressing restlessness and need to move often accompanied by an inability to sit or stand still. This is most likely to occur within the first few weeks of treatment. In patients who

develop these symptoms, increasing the dose may be detrimental and it may be necessary to review the use of fluvoxamine.

Withdrawal symptoms seen on discontinuation of fluvoxamine treatment: Withdrawal symptoms when treatment is discontinued are common, particularly if discontinuation is abrupt. The risk of withdrawal symptoms may be dependent on several factors including the duration and dose of therapy and the rate of dose reduction.

Dizziness, sensory disturbance (including paraesthesia, visual disturbances and electric shock sensations), sleep disturbances (including insomnia and intense dreams), agitation and anxiety, irritability, confusion, emotional instability, nausea and/or vomiting and diarrhoea, sweating and palpitations, headache and tremor are the most commonly reported reactions. Generally, these symptoms are mild to moderate, however, in some patients they may be severe in intensity. They usually occur within the first few days of discontinuing treatment, but there have been very rare reports of such symptoms in patients who have inadvertently missed a dose. Generally these symptoms are self-limiting and usually resolve within 2 weeks, though in some individuals they may be prolonged (2-3 months or more). It is therefore advised that fluvoxamine treatment should be gradually discontinued over a period of several weeks or months, according to the patient's needs.

Patients suffering from hepatic and renal insufficiency should start on a low dose and be carefully monitored. Treatment with fluvoxamine has rarely increases liver enzymes, generally associated with symptoms. In such cases, treatment should be discontinued.

Glycaemic control may be disturbed, especially in the early stages of treatment. The dosage of anti-diabetic drugs may need to be adjusted.

Caution is recommended when the drug is administered to patients with unstable epilepsy and patients with controlled epilepsy should be carefully monitored. Treatment with fluvoxamine should be discontinued if seizures occur or if seizure frequency increases.

On rare occasions development of a serotonin syndrome or neuroleptic malignant syndrome-like events have been reported in association with treatment of fluvoxamine, particularly when given combination with other serotonergic and/or neuroleptic drugs. As these syndromes may result in

potentially life-threatening conditions, treatment with fluvoxamine should be discontinued if such events occur and supportive treatment should be initiated.

As with other SSRIs, hyponatremia has been rarely reported, and appears to be reversible when fluvoxamine is discontinued. Some cases were possibly due to the syndrome of inappropriate antidiuretic hormone secretion. The majority of reports were associated with older patients.

There have been reports of cutaneous bleeding abnormalities such as ecchymoses and purpura with SSRIs. Caution is advised in patients taking SSRIs particularly in concomitant use with drugs known to affect platelet function as well as in patients with a history of bleeding or coagulation disorders. Fluvoxamine should be used with caution in patients with a history of mania/hypomania. Fluvoxamine should be discontinued in any patient entering a manic phase.

When combined with fluvoxamine, plasma concentration of terfenadine, astemizole or cisapride may be increased resulting in an increased risk for QT-prolongation/Torsade de Pointes. Therefore, fluvoxamine should not be co-administered with these drugs.

Due to lack of clinical experience, special attention is advised in the situation of post-acute myocardial infarction.

There is limited clinical experience of concomitant administration of fluvoxamine and ECT therefore caution is advisable.

Data in elderly subjects give no indication of clinically significant differences in normal daily dosages compared to younger subjects. However upward dose titration should be done slower in the elderly, and dosing should always be done with caution.

Suicidality in Children and Adolescents:

- Antidepressants increase the risk of suicidal thinking and behavior (suicidality) in children and adolescents with major depressive disorder (MDD) and other psychiatric disorders.
- Anyone considering the use of an antidepressant in a child or adolescent for any clinical use must balance the risk of increased suicidality with the clinical need.
- Patients who are started on therapy should be observed closely for clinical worsening, suicidality, or unusual changes in behavior.

- Families and caregivers should be advised to closely observe the patient and to communicate with the prescriber.

Sexual dysfunction: Selective serotonin reuptake inhibitors (SSRIs)/serotonin norepinephrine reuptake inhibitors (SNRIs) may cause symptoms of sexual dysfunction.

There have been reports of long-lasting sexual dysfunction where the symptoms have continued despite discontinuation of SSRIs/SNRIs.

Use in Pregnancy and Lactation:

Safe use of fluvoxamine during pregnancy and lactation has not been established. Neonates exposed to antipsychotic drugs during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, hypotonia, tremor, somnolence, respiratory distress, and feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self-limited, in other cases neonates have required intensive care unit support and prolonged hospitalisation. Therefore, it should not be administered to women of childbearing potential or nursing mothers unless, in the opinion of the treating physician, the expected benefits to the patient outweigh the possible hazards to the child or fetus.

Observational data indicate an increased risk (less than 2-fold) of postpartum haemorrhage following SSRI/SNRI exposure within the month prior to birth.

Drug Interactions

Combined use of APO-FLUVOXAMINE (fluvoxamine maleate) and MAO inhibitors is contraindicated (see Contraindications).

An increase in previously stable plasma levels of those tricyclic antidepressants (e.g., clomipramine, imipramine, amitriptyline) and neuroleptics (e.g., clozapine, olanzapine) which are largely metabolised through cytochrome P450 1A2 when given together with fluvoxamine, has been reported. A decrease in the dose of these products should be considered if treatment with fluvoxamine is initiated.

Patients co-administered fluvoxamine and CYP1A2 metabolised drugs with a narrow therapeutic index (such as tacrine, theophylline, methadone, mexiletine) should be carefully monitored and, if necessary, dose adjustment of these drugs is recommended.

When given with fluvoxamine, warfarin plasma concentrations were significantly increased and prothrombin times prolonged. As plasma concentrations of propranolol are increased in combination with fluvoxamine, the propranolol dose may need to be lowered.

Caffeine plasma levels are likely to be increased during coadministration with fluvoxamine. Thus, patients who consume high quantities of caffeine-containing beverages should lower their intake when fluvoxamine is administered and adverse caffeine effects (like tremor, palpitations, nausea, restlessness, insomnia) are observed.

As plasma concentrations of ropinirole may be increased in combination with Fluvoxamine thus increasing the risk of overdose, surveillance and reduction in the posology of ropinirole during fluvoxamine treatment and after its withdrawal may be required.

Patients co-administered fluvoxamine and CYP2C metabolised drugs with a narrow therapeutic index (such as phenytoin) should be carefully monitored and, if necessary, dose adjustment of these drugs is recommended. Patients co-administered fluvoxamine and CYP3A4 metabolised drugs with a narrow therapeutic index (such as carbamazepine, ciclosporin) should be carefully monitored and, if necessary, dose adjustment of these drugs is recommended.

The plasma levels of oxidatively metabolised benzodiazepines (e.g. triazolam, midazolam, alprazolam, and diazepam) are likely to be increased when co-administered with fluvoxamine. The dosage of these benzodiazepines should be reduced during coadministration with fluvoxamine. Fluvoxamine does not influence plasma concentrations of digoxin. Fluvoxamine does not influence plasma concentrations of atenolol. The serotonergic effects of fluvoxamine may be enhanced when used in combination with other serotonergic agents (including triptans, SSRIs and St. John's Wort preparations). (See Special warnings and precautions for use).

Fluvoxamine has been used in combination with lithium in the treatment of severely ill, drug-resistant patients. However, lithium (and possibly also tryptophan) enhances the serotonergic effects of fluvoxamine. The combination should be used with caution in patients with severe, drug-resistant depression. In patients on oral anticoagulants and fluvoxamine, the risk for haemorrhage may increase and these patients should therefore be closely monitored.

As with other psychotropic drugs patients should be advised to avoid alcohol use while taking fluvoxamine.

Adverse Effects

Nausea, sometimes accompanied by vomiting is the most frequently observed symptom associated with fluvoxamine treatment. This side effect usually diminishes within the first two weeks of treatment. Other adverse events, observed in clinical studies at frequencies listed below, are often associated with the illness and are not necessarily related to treatment.

Common (Frequency 1-10%):

Body: Asthenia, headache, malaise

Cardiovascular: Palpitations/tachycardia

Digestive system: Abdominal pain, anorexia, constipation, diarrhoea, dry mouth, dyspepsia

Nervous system: Agitation, anxiety, dizziness, insomnia, nervousness, somnolence, tremor.

Skin: Sweating

Uncommon (Frequency <1%):

Cardiovascular: (Postural) hypotension

Musculoskeletal: Arthralgia, myalgia

Nervous System: Ataxia, confusion, extrapyramidal symptoms, hallucinations

Urogenital: Abnormal (delayed) ejaculation

Skin: Cutaneous hypersensitivity reactions (incl. rash, pruritus, angioedema).

Rare (Frequency <0.1%):

Digestive system: Liver function abnormality

Nervous system: Convulsions, mania

Urogenital: Galactorrhoea

Skin: Photosensitivity

Psychomotor restlessness/akathisia (see section 4.4 Special warnings and precautions for use).

Other adverse events observed during marketing:

Weight gain and loss have been reported.

Rarely, serotonin syndrome, neuroleptic malignant syndrome-like events, hyponatremia

and SIADH have been reported (see also 4.4 Special Warnings and precautions for use).

It is possible that withdrawal reactions may occur on stopping therapy with fluvoxamine although the available preclinical and clinical evidence does not suggest that this treatment causes dependence. The following symptoms have been reported in association with withdrawal of the product: dizziness, paresthesia, headache, nausea and anxiety. The majority of the withdrawal reactions are mild and self-limiting.

When stopping, a gradual dose reduction may be considered.

Haemorrhage: (see warnings and precautions for use)

Very rarely, paresthesia, anorgasmia and taste perversion have been reported.

Symptoms and Treatment of Overdose

Symptoms: Symptoms include gastro-intestinal complaints (nausea, vomiting and diarrhoea), somnolence and dizziness. Cardiac events (tachycardia, bradycardia, hypotension), liver function disturbances, convulsions and coma have also been reported.

Fluvoxamine has a wide margin of safety in overdose. Since market introduction, reports of deaths attributed to overdose of fluvoxamine alone have been extremely rare. The highest documented dose of fluvoxamine ingested by a patient is 12 grams. This patient recovered completely. Occasionally, more serious complications were observed in cases of deliberate overdose of fluvoxamine in combination with other drugs.

Treatment: There is no specific antidote to fluvoxamine. In case of overdose the stomach should be emptied as soon as possible after tablet ingestion and symptomatic treatment should be given. The repeated use of medicinal charcoal, if necessary accompanied by an osmotic laxative, is also recommended. Forced diuresis or dialysis are unlikely to be of benefit.

Product Description/Availability of Dosage Forms

Each round, white biconvex, film-coated tablet, engraved "APO" on one side and "50" on the other, contains fluvoxamine maleate 50 mg. Available in HDPE bottles of 100's.

Storage Conditions

Preserve in well-closed containers. Store in a dry place below 30°C.

Shelf-life

18 months from the date of manufacture.

Manufacturer

Apotex Inc., 150 Signet Drive, Toronto Ontario, Canada M9L 1T9.

Date of Revision of Package Insert

21st Oct 2025

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CA 0079C